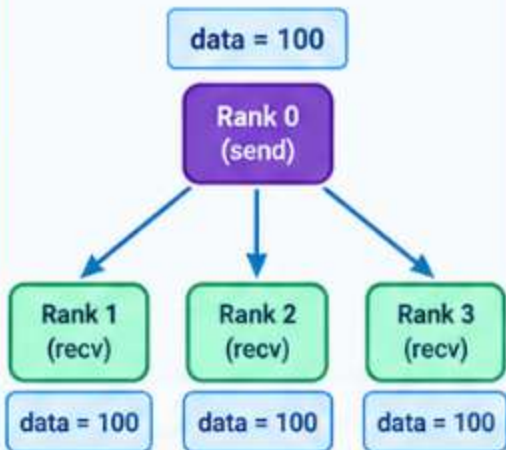


MPI_Bcast

Broadcasts the same data from one process to all processes.

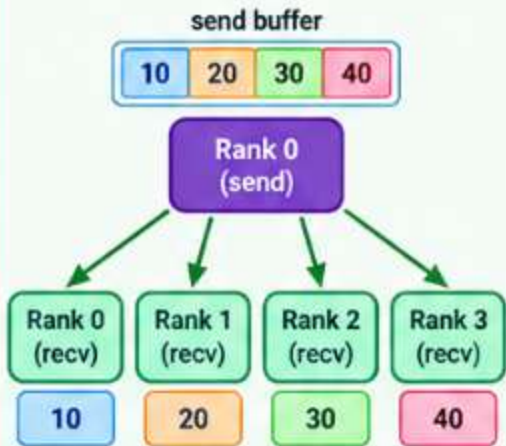


Key idea:

One process sends the same data to all processes in the communicator.

MPI_Scatter

Distributes different data from one process to all processes.

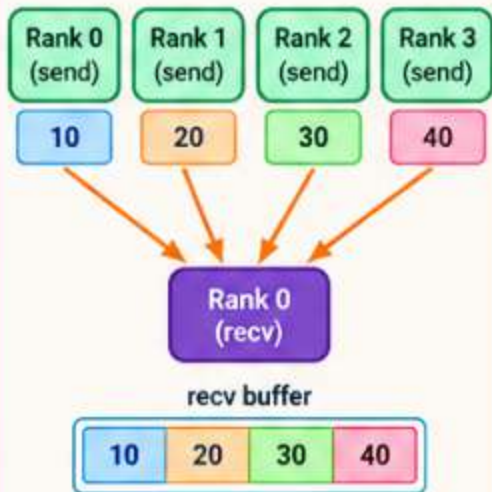


Key idea:

One process sends different pieces of data to each process.

MPI_Gather

Collects data from all processes and sends it to one process.

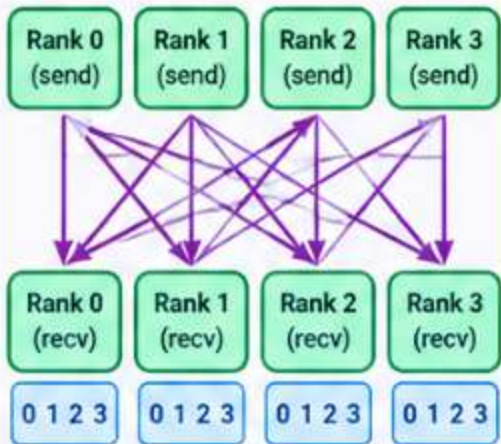


Key idea:

All processes send their data to one process (root).

MPI_Allgather

Each process sends data to all processes, and each process receives data from all processes.

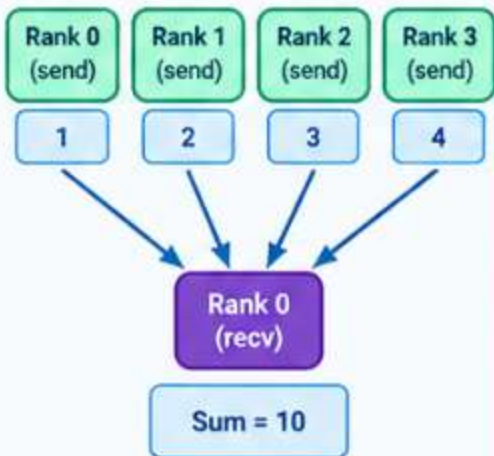


Key idea:

All processes send their data to all processes, and all processes receive the data from all processes.

MPI_Reduce

Combines data from all processes using a reduction operation and sends the result to one process.

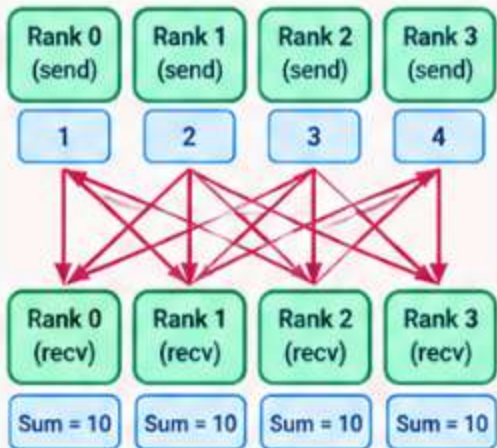


Key idea:

All processes contribute data, which is reduced (e.g., sum, max, etc.) and sent to one process (root).

MPI_Allreduce

Performs a reduction operation and gives the result to all processes.



Key idea:

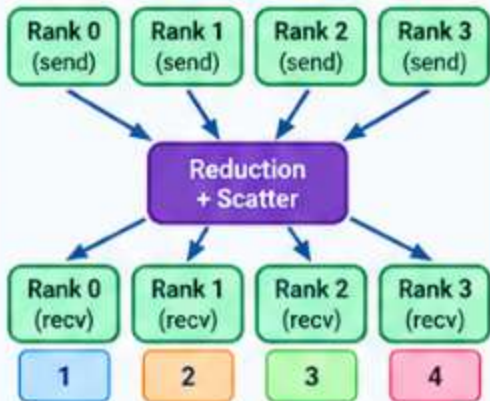
All processes contribute data, perform the reduction, and all processes receive the result.



MPI_Reduce_scatter

Reduces data across all processes and distributes different portions of the result to each process.

send buffer (example)



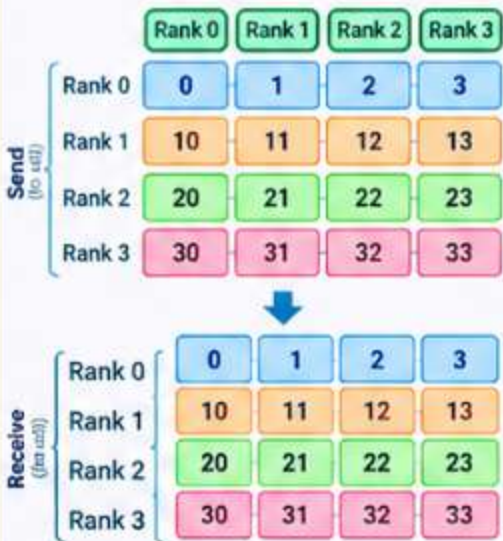
Key idea:

First reduces data across all processes, then scatters different parts of the result to each process.



MPI_AlltoAll

Each process sends different data to every other process, and receives data from every other process.



Key idea:

Every process sends a different piece of data to every other process and receives data from every other process.